

Lab 3 – Architecture Selection & Justification

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Problem Statement: Hyperlocal Courier Dispatch & Tracking Engine

Architecture: Microservices Architecture

Architecture Selection

We chose Microservices Architecture for the Hyperlocal Courier Dispatch & Tracking Engine.

Architectural Choice: The system is divided into independently functioning services such as the Courier Request Service, Rider Assignment & Dispatch Service, Delivery Tracking Service, Route Optimization Service, and Delivery Verification Service. This structure matches the different responsibilities of the courier system.

Reason 1: Microservices allows the rider assignment and dispatch functionality to operate independently from other services. This is useful because the system must find the nearest active rider within a 3 km radius and send a dispatch notification to that rider.

Reason 2: The architecture is suitable for the system's real-time tracking and route optimization requirements. The Delivery Tracking Service can handle GPS updates independently, while the Route Optimization Service can process multi-stop routes without requiring the entire system to be modified.

Security Advantage: Separating services limits access between different parts of the system. In particular, delivery verification can be isolated as a separate service responsible for OTP verification before a delivery is marked as completed.

Performance Benefit: Individual services can handle their workloads independently. The Delivery Tracking Service can therefore focus on GPS and delivery-status updates, helping the system meet the requirement of delivering tracking updates with under 2-second latency.